

WASP-36b

R-2202

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_140214 · created 2026-08-14T00:50:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456702.87474 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.023
Transit depth (Rp/Rs)^2	1.87 +/- 0.63 [%]
Orbital Inclination (inc)	84.79 +/- 1.47
Airmass coefficient 1 (a1)	1.021 +/- 0.022
Airmass coefficient 2 (a2)	-0.0097 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	2.14 %
Best Comparison Star	#3 - [522, 431]
Optimal Method	PSF photometry
Transit Duration (day)	0.0838 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.137 ± 0.023 vs 0.1368 (deviation 0.01σ)
Duration fit vs published	0.0838 d vs 0.0758 d (deviation 0.87σ)
Mid-time O–C	-2.3 min
Depth SNR	4.9
Residual scatter	2.14 %

Light curve

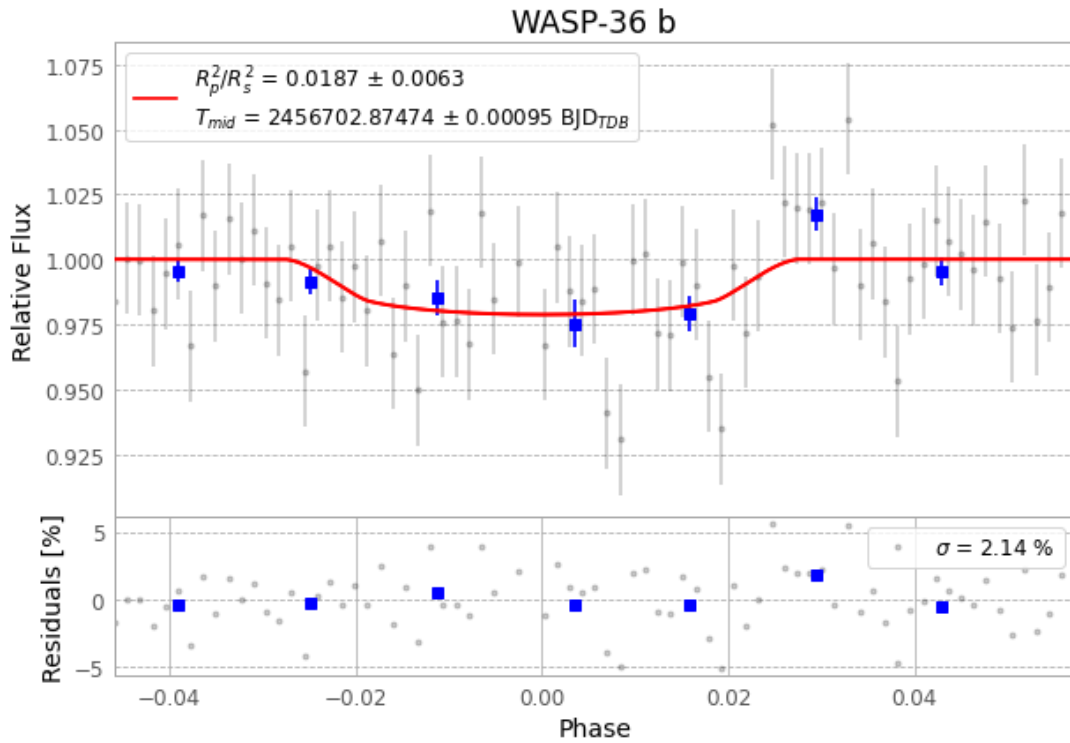


Figure 1 — FinalLightCurve_WASP-36 b_2014-02-14.png (SHA-256 1f571c15940d86e9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6478ca3dedfa4434...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

77 FITS frames (51 MB), WASP-36140214070913.FITS → WASP-36140214105712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140214.log (ae0d9149bd4e...), FinalLightCurve_WASP-36 b_2014-02-14.png (1f571c15940d...), FinalLightCurve_WASP-36 b_2014-02-14.csv (ece778621f14...)

Compute resources

Wall time 122.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:50Z.